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TREND-BASED MOCK PAPER

GATE Geology and Geophysics

Mock Paper — 04

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Geology and Geophysics
Paper:	Mock 04
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

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Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section A: General Aptitude

Q1. [1-MCQ] [1 mark]

Choose similar to 'volcanic':

- (A) Gentle
- (B) Explosive
- (C) Calm
- (D) Peaceful

Section B: Geology and Geophysics

Q2. [1-MCQ] [1 mark]

Mean of 1,2,3,4,5:

- (A) 2
- (B) 3
- (C) 4
- (D) 5

Q3. [1-MCQ] [1 mark]

$d/dx[\ln(\sec x) + \ln(\tan x)]$ evaluated at $x = \pi/4$ equals:

- (A) 0
- (B) 2
- (C) 4
- (D) 1

Q4. [1-MCQ] [1 mark]

For matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $A^{50} \cdot [1, 0]^T$ equals:

- (A) $[1, 0]^T$
- (B) $[2, 1]^T$
- (C) $[(3+2\sqrt{2})^{50}, \dots]$

(D) Cannot determine

Q5. [1-MCQ] [1 mark]

For the ODE $dy/dx = y/x + \sqrt{y^2/x^2}$, with $y(1)=0$, the solution is:

- (A) $y = 0$
- (B) $y = x \ln(x)$
- (C) $y = x \sin(\ln x)$
- (D) $y = x^2$

Q6. [1-MCQ] [1 mark]

The rank of the matrix $\begin{bmatrix} 1,2,3 \\ 2,4,6 \\ 3,6,9 \end{bmatrix}$ is:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q7. [1-MCQ] [1 mark]

Laplace transform of $e^{(-3t)}u(t)$ is:

- (A) $1/(s+3)$
- (B) $1/(s-3)$
- (C) $1/s$
- (D) e^{-3}/s

Q8. [1-NAT] [1 mark]

If $f(z) = z^2 + 3z + 2$ is analytic, then $\operatorname{Re}(f(i)) =$

(Numerical Answer Type — Enter the numerical value)

Q9. [1-MCQ] [1 mark]

Fourier series of $f(x)=x$ on $(-\pi,\pi)$ has only:

- (A) Sine terms
- (B) Cosine terms
- (C) Both
- (D) Neither

Q10. [1-NAT] [1 mark]

The area of a circle of radius r is:

(Numerical Answer Type — Enter the numerical value)

Q11. [1-NAT] [1 mark]

Rolle's theorem requires $f(x)$ to be:

(Numerical Answer Type — Enter the numerical value)

Q12. [1-MCQ] [1 mark]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

- (A) 0.2
- (B) 0.5
- (C) 0.6
- (D) 0.4

Q13. [1-MCQ] [1 mark]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

- (A) e
- (B) e^2
- (C) 1
- (D) infinity

Q14. [1-NAT] [1 mark]

The vector (1,2,3) projected onto (2,1,0) is:

(Numerical Answer Type — Enter the numerical value)

Q15. [1-MCQ] [1 mark]

If $f(x) = x^3 - 3x + 2$, number of real roots:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q16. [1-NAT] [1 mark]

Which of the following is a probability generating function for Poisson(λ)?

(Numerical Answer Type — Enter the numerical value)

Q17. [1-NAT] [1 mark]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

(Numerical Answer Type — Enter the numerical value)

Q18. [1-MCQ] [1 mark]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\text{trace}(A^2)$ equals:

- (A) 14
- (B) 6
- (C) 36
- (D) 12

Q19. [1-MCQ] [1 mark]

Number of spanning trees in complete graph K_4 :

- (A) 4
- (B) 8
- (C) 16
- (D) 12

Q20. [1-MCQ] [1 mark]

Gauss-Seidel method converges when the coefficient matrix is:

- (A) Symmetric
- (B) Positive definite
- (C) Diagonally dominant
- (D) Non-singular

Q21. [2-MCQ] [2 marks]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq$:

- (A) $\|u\| * \|v\|$

- (B) $\|u\| + \|v\|$
- (C) $\|u\| / \|v\|$
- (D) $\sqrt{\|u\| * \|v\|}$

Q22. [2-NAT] [2 marks]

The differential equation $y \, dx - x \, dy = 0$ has solution:

(Numerical Answer Type — Enter the numerical value)

Q23. [2-MCQ] [2 marks]

Consider $f(x) = |x|$ on $[-1,1]$. This function is:

- (A) Differentiable everywhere
- (B) Continuous but not differentiable at 0
- (C) Neither continuous nor differentiable
- (D) Differentiable at 0 only

Q24. [2-MCQ] [2 marks]

$P(|Z| \leq 1)$ for $Z \sim N(0,1)$ is approximately:

- (A) 0.68
- (B) 0.32
- (C) 0.95
- (D) 0.16

Q25. [2-MCQ] [2 marks]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

- (A) 0.908
- (B) 0.092
- (C) 0.762
- (D) 0.195

Q26. [2-MSQ] [2 marks]

The Laplace transform of $t^2 e^{-2t}$ is:

- (A) $1/(s+2)^2$
- (B) $1/(s-2)^2$
- (C) $s/(s+2)^2$
- (D) $2/(s+2)^2$

Q27. [2-MCQ] [2 marks]

For a Poisson distribution with mean 4, variance is:

- (A) 4
- (B) 2
- (C) 16
- (D) 1

Q28. [2-NAT] [2 marks]

Laplace transform of $e^{-3t} u(t)$ is:

(Numerical Answer Type — Enter the numerical value)

Q29. [2-MCQ] [2 marks]

Which of the following is a probability generating function for $\text{Poisson}(\lambda)$?

- (A) $e^{\lambda(z-1)}$
- (B) $1/(1-z)$

- (C) $\ln(1-z)$
- (D) $e^{(-\lambda)(1+z)}$

Q30. [2-NAT] [2 marks]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

(Numerical Answer Type — Enter the numerical value)

Q31. [2-MCQ] [2 marks]

The area of a circle of radius r is:

- (A) πr^2
- (B) $2\pi r$
- (C) πr
- (D) $2\pi r^2$

Q32. [2-MCQ] [2 marks]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq$:

- (A) $\|u\| * \|v\|$
- (B) $\|u\| + \|v\|$
- (C) $\|u\| / \|v\|$
- (D) $\sqrt{\|u\| * \|v\|}$

Q33. [2-MCQ] [2 marks]

Which of the following is a probability generating function for $\text{Poisson}(\lambda)$?

- (A) $e^{\lambda(z-1)}$
- (B) $1/(1-z)$
- (C) $\ln(1-z)$
- (D) $e^{(-\lambda)(1+z)}$

Q34. [2-MCQ] [2 marks]

Number of spanning trees in complete graph K_4 :

- (A) 4
- (B) 8
- (C) 16
- (D) 12

Q35. [2-MCQ] [2 marks]

The differential equation $y \, dx - x \, dy = 0$ has solution:

- (A) $y = Cx$
- (B) $y = C/x$
- (C) $y = Cx^2$
- (D) $x^2 + y^2 = C$

Q36. [2-MCQ] [2 marks]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

- (A) 1
- (B) 0
- (C) -1
- (D) $\cos(\pi)$

Q37. [2-MCQ] [2 marks]

If $f(z) = z^2 + 3z + 2$ is analytic, then $\operatorname{Re}(f(i)) =$

- (A) -1
- (B) -2
- (C) 1
- (D) 2

Q38. [2-MCQ] [2 marks]

Fourier series of $f(x)=x$ on $(-\pi,\pi)$ has only:

- (A) Sine terms
- (B) Cosine terms
- (C) Both
- (D) Neither

Q39. [2-MCQ] [2 marks]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

- (A) 0.2
- (B) 0.5
- (C) 0.6
- (D) 0.4

Q40. [2-MCQ] [2 marks]

The differential equation $y \, dx - x \, dy = 0$ has solution:

- (A) $y = C \cdot x$
- (B) $y = C/x$
- (C) $y = C \cdot x^2$
- (D) $x^2 + y^2 = C$

Q41. [2-NAT] [2 marks]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\operatorname{trace}(A^2)$ equals:

(Numerical Answer Type — Enter the numerical value)

Q42. [2-MCQ] [2 marks]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq$:

- (A) $\|u\| \cdot \|v\|$
- (B) $\|u\| + \|v\|$
- (C) $\|u\| / \|v\|$
- (D) $\sqrt{\|u\| \cdot \|v\|}$

Q43. [2-NAT] [2 marks]

Rolle's theorem requires $f(x)$ to be:

(Numerical Answer Type — Enter the numerical value)

Q44. [1-MCQ] [1 mark]

Study of fossils:

- (A) Paleontology
- (B) Petrology
- (C) Mineralogy
- (D) None

Q45. [1-NAT] [1 mark]

Hardest mineral:

(Numerical Answer Type — Enter the numerical value)

Q46. [1-MCQ] [1 mark]

Law of superposition by:

- (A) Steno
 - (B) Lyell
 - (C) Darwin
 - (D) Hutton
-

Q47. [1-MCQ] [1 mark]

Igneous rock from lava:

- (A) Intrusive
 - (B) Extrusive
 - (C) Metamorphic
 - (D) Sedimentary
-

Q48. [1-MCQ] [1 mark]

Hardest mineral on Mohs scale:

- (A) Corundum
 - (B) Diamond
 - (C) Topaz
 - (D) Quartz
-

Q49. [2-NAT] [2 marks]

Igneous rock from lava:

(Numerical Answer Type — Enter the numerical value)

Q50. [2-MSQ] [2 marks]

Hardest mineral on Mohs scale:

- (A) Corundum
 - (B) Diamond
 - (C) Topaz
 - (D) Quartz
-

Q51. [2-NAT] [2 marks]

Hardest mineral on Mohs scale:

(Numerical Answer Type — Enter the numerical value)

Q52. [2-MSQ] [2 marks]

Igneous rock from lava:

- (A) Intrusive
 - (B) Extrusive
 - (C) Metamorphic
 - (D) Sedimentary
-

Q53. [2-MSQ] [2 marks]

Igneous rock from lava:

- (A) Intrusive

- (B) Extrusive
- (C) Metamorphic
- (D) Sedimentary

Q54. [2-MCQ] [2 marks]

Hardest mineral on Mohs scale:

- (A) Corundum
- (B) Diamond
- (C) Topaz
- (D) Quartz

Q55. [2-MCQ] [2 marks]

Hardest mineral on Mohs scale:

- (A) Corundum
- (B) Diamond
- (C) Topaz
- (D) Quartz

Q56. [2-MCQ] [2 marks]

Igneous rock from lava:

- (A) Intrusive
- (B) Extrusive
- (C) Metamorphic
- (D) Sedimentary

Q57. [2-MCQ] [2 marks]

Hardest mineral on Mohs scale:

- (A) Corundum
- (B) Diamond
- (C) Topaz
- (D) Quartz

Q58. [2-NAT] [2 marks]

Igneous rock from lava:

(Numerical Answer Type — Enter the numerical value)

Q59. [2-NAT] [2 marks]

Igneous rock from lava:

(Numerical Answer Type — Enter the numerical value)

Q60. [2-MCQ] [2 marks]

Hardest mineral on Mohs scale:

- (A) Corundum
- (B) Diamond
- (C) Topaz
- (D) Quartz

Q61. [1-NAT] [1 mark]

P-waves are:

(Numerical Answer Type — Enter the numerical value)

Q62. [1-MCQ] [1 mark]

Gravity anomaly due to:

- (A) Density variation
- (B) Magnetism
- (C) Electricity
- (D) None

Q63. [1-MCQ] [1 mark]

Earth's magnetic field generated by:

- (A) Core
- (B) Crust
- (C) Mantle
- (D) Atmosphere

Q64. [1-MCQ] [1 mark]

Landsat uses:

- (A) Optical
- (B) Radar
- (C) Both
- (D) None

Q65. [1-MCQ] [1 mark]

Igneous rock from:

- (A) Sediment
 - (B) Cooling magma
 - (C) Metamorphism
 - (D) Organic
-

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1MCQ	B	1
2	1MCQ	B	1
3	1MCQ	A	1
4	1NAT	B	1
5	1MCQ	A	1
6	1NAT	A	1
7	1MCQ	A	1
8	1MCQ	A	1
9	1MCQ	A	1
10	1MCQ	B	1
11	1MCQ	B	1
12	1MCQ	B	1
13	1MCQ	C	1

14	1MCQ	A	1
15	1MCQ	A	1
16	1MCQ	A	1
17	1MCQ	A	1
18	1NAT	A	1
19	1MCQ	A	1
20	1NAT	A	1
21	1NAT	C	1
22	1MCQ	B	1
23	1MCQ	B	1
24	1NAT	B	1
25	1MCQ	B	1
26	1NAT	A	1
27	1NAT	C	1
28	1MCQ	A	1

29	1MCQ	C	1
30	1MCQ	C	1
31	2MCQ	A	2
32	2NAT	B	2
33	2MCQ	B	2
34	2MCQ	A	2
35	2MCQ	A	2
36	2MSQ	A	2
37	2MCQ	A	2
38	2NAT	A	2
39	2NAT	B	2
40	2MSQ	B	2
41	2MCQ	A	2
42	2NAT	A	2
43	2NAT	B	2

44	2MSQ	B	2
45	2MCQ	A	2
46	2MCQ	A	2
47	2MCQ	A	2
48	2MSQ	B	2
49	2MCQ	B	2
50	2MCQ	C	2
51	2MCQ	B	2
52	2MCQ	C	2
53	2MCQ	A	2
54	2MCQ	A	2
55	2MCQ	B	2
56	2MCQ	B	2
57	2MCQ	B	2
58	2MCQ	B	2

59	2NAT	B	2
60	2MCQ	B	2
61	2NAT	A	2
62	2MCQ	A	2
63	2NAT	C	2
64	2NAT	B	2
65	2MCQ	B	2

DETAILED SOLUTIONS

Q1.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Verbal | Difficulty: easy

Q2.

Answer: B

Solution:

Correct Answer: B

Subject: Engineering Mathematics | Topic: Statistics | Difficulty: moderate

Q3.

Answer: A

Solution:

Correct Answer: A

Subject: Geology | Topic: Fossils | Difficulty: moderate

Q4.

Answer: B

Solution:

Correct Answer: B

Subject: Geology | Topic: Minerals | Difficulty: moderate

Q5.

Answer: A

Solution:

Correct Answer: A

Subject: Geology | Topic: Stratigraphy | Difficulty: easy

Q6.

Answer: A

Solution:

Correct Answer: A

Subject: Geophysics | Topic: Seismic | Difficulty: easy

Q7.

Answer: A

Solution:

Correct Answer: A

Subject: Geophysics | Topic: Gravity | Difficulty: moderate

Q8.

Answer: A

Solution:

Correct Answer: A

Subject: Geophysics | Topic: Magnetism | Difficulty: easy

Q9.

Answer: A

Solution:

Correct Answer: A

Subject: Remote Sensing & GIS | Topic: Satellite | Difficulty: moderate

Q10.

Answer: B

Solution:

Correct Answer: B

Subject: Petrology | Topic: Rocks | Difficulty: easy

Q11.

Answer: B

Solution:

Extrusive (volcanic): from lava at surface. Intrusive: magma below surface.

Subject: Geology | Topic: Geology | Difficulty: easy

Q12.

Answer: B

Solution:

Mohs: Talc(1)..Quartz(7)..Topaz(8)..Corundum(9)..Diamond(10). Diamond hardest.

Subject: Geology | Topic: Geology | Difficulty: moderate

Q13.

Answer: C

Solution:

$\frac{d}{dx}[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm. Let me recalc: $\frac{d}{dx} \ln(\sec x) = \tan x$. $\frac{d}{dx} \ln(\tan x) = \sec^2 x / \tan x = 1/(\sin x \cos x) = 2\csc(2x)$. At $\pi/4$: $1 + 2 = 3$. Not in options. The derivative of $\ln(\sec x * \tan x) = \frac{d}{dx}[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1+2=3$. Hmm.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q14.

Answer: A

Solution:

Eigenvalues are 3 and 1. $[1,0]^T$ is eigenvector for eigenvalue 1. $A^{50}[1,0]^T = [1,0]^T$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q15.

Answer: A

Solution:

Homogeneous ODE. Substituting $y=x*v$ and $y(1)=0$ gives $y=0$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q16.

Answer: A

Solution:

Row 2 = 2*Row 1, Row 3 = 3*Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q17.

Answer: A

Solution:

$L\{e^{-at}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q18.

Answer: A

Solution:

$f(i) = i^2 + 3i + 2 = -1 + 3i$. $\text{Re}(f(i)) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q19.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q20.

Answer: A

Solution:

Area = πr^2 . Circumference = $2\pi r$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q21.

Answer: C

Solution:

Rolle's: f continuous on $[a,b]$, differentiable on (a,b) , $f(a)=f(b)$. Then $f'(c)=0$ for some c in (a,b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q22.

Answer: B

Solution:

$$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q23.

Answer: B

Solution:

$$(1+1/n)^{2n} = [(1+1/n)^n]^2 \rightarrow e^2 \text{ as } n \rightarrow \infty.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q24.

Answer: B

Solution:

$$\text{proj} = ((v \cdot w)/(w \cdot w)) * w = (4/5) * (2, 1, 0) = (8/5, 4/5, 0).$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q25.

Answer: B

Solution:

$$f(0)=2, f(1)=0, f(-1)=4, f(2)=4. f(-2)=-8+6+2=0. \text{ Roots at } x=1 \text{ (double?) and } x=-2. f(x)=(x-1)^2(x+2). \text{ Two distinct real roots.}$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q26.

Answer: A

Solution:

$$\text{PGF of Poisson}(\lambda): G(z) = E[z^X] = \exp(\lambda(z-1)).$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q27.

Answer: C

Solution:

$$y = A \cos(2x) + B \sin(2x). A=1, B=0. y(\pi/2) = \cos(\pi) = -1.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q28.

Answer: A

Solution:

$$\text{Eigenvalues of } A^2 \text{ are } 1, 4, 9. \text{ Trace} = \text{sum} = 1+4+9 = 14.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q29.

Answer: C

Solution:

$$\text{Kirchhoff: for } K_n, \text{ spanning trees} = n^{n-2} = 4^2 = 16.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q30.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q31.

Answer: A

Solution:

$$|\langle u, v \rangle| \leq \|u\| * \|v\|. \text{ Equality when } u \text{ and } v \text{ are linearly dependent.}$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q32.

Answer: B

Solution:

$$dy/y = dx/x. \ln(y) = \ln(x) + C. y = C*x. \text{ Wait: } y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C*x. \text{ That's option A. Hmm, let me check: } y \, dx - x \, dy = 0 \Rightarrow y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow y = C*x. \text{ Answer: A.}$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q33.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q34.

Answer: A

Solution:

Standard normal: $P(-1 < Z < 1)$ approx 0.68.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q35.

Answer: A

Solution:

$P(X \geq 2) = 1 - e^{-4(1+4)} = 1 - 5e^{-4}$ approx 0.908.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q36.

Answer: A

Solution:

$L\{t^2 f(t)\} = -d/ds(F(s))$. $F(s) = 1/(s+2)$. $-d/ds[1/(s+2)] = 1/(s+2)^2$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q37.

Answer: A

Solution:

Poisson: mean = λ = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q38.

Answer: A

Solution:

$L\{e^{-at}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q39.

Answer: B

Solution:

Extrusive (volcanic): from lava at surface. Intrusive: magma below surface.

Subject: Geology | Topic: Geology | Difficulty: easy

Q40.

Answer: B

Solution:

Mohs: Talc(1)..Quartz(7)..Topaz(8)..Corundum(9)..Diamond(10). Diamond hardest.

Subject: Geology | Topic: Geology | Difficulty: moderate

Q41.

Answer: A

Solution:

PGF of Poisson(λ): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q42.

Answer: A

Solution:

$P(X \geq 2) = 1 - e^{-4(1+4)} = 1 - 5e^{-4}$ approx 0.908.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q43.

Answer: B

Solution:

Mohs: Talc(1)..Quartz(7)..Topaz(8)..Corundum(9)..Diamond(10). Diamond hardest.

Subject: Geology | Topic: Geology | Difficulty: easy

Q44.

Answer: B

Solution:

Extrusive (volcanic): from lava at surface. Intrusive: magma below surface.

Subject: Geology | Topic: Geology | Difficulty: moderate

Q45.

Answer: A

Solution:

Area = πr^2 . Circumference = $2\pi r$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q46.

Answer: A

Solution:

$|\langle u, v \rangle| \leq \|u\| * \|v\|$. Equality when u and v are linearly dependent.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q47.

Answer: A

Solution:

PGF of Poisson(λ): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q48.

Answer: B

Solution:

Extrusive (volcanic): from lava at surface. Intrusive: magma below surface.

Subject: Geology | Topic: Geology | Difficulty: moderate

Q49.

Answer: B

Solution:

Mohs: Talc(1)..Quartz(7)..Topaz(8)..Corundum(9)..Diamond(10). Diamond hardest.

Subject: Geology | Topic: Geology | Difficulty: easy

Q50.

Answer: C

Solution:

Kirchhoff: for K_n , spanning trees = $n^{n-2} = 4^2 = 16$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q51.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C*x$. Wait: $y dx = x dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C*x$. That's option A. Hmm, let me check: $y dx - x dy = 0 \Rightarrow y dx = x dy \Rightarrow dy/y = dx/x \Rightarrow y = C*x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q52.

Answer: C

Solution:

$y = A*\cos(2x) + B*\sin(2x)$. $A=1$, $B=0$. $y(\pi/2) = \cos(\pi) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q53.

Answer: A

Solution:

$f(i) = i^2 + 3i + 2 = -1 + 3i$. $\text{Re}(f(i)) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q54.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q55.

Answer: B

Solution:

$$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q56.

Answer: B

Solution:

Mohs: Talc(1)..Quartz(7)..Topaz(8)..Corundum(9)..Diamond(10). Diamond hardest.

Subject: Geology | Topic: Geology | Difficulty: moderate

Q57.

Answer: B

Solution:

Extrusive (volcanic): from lava at surface. Intrusive: magma below surface.

Subject: Geology | Topic: Geology | Difficulty: easy

Q58.

Answer: B

Solution:

Mohs: Talc(1)..Quartz(7)..Topaz(8)..Corundum(9)..Diamond(10). Diamond hardest.

Subject: Geology | Topic: Geology | Difficulty: moderate

Q59.

Answer: B

Solution:

Extrusive (volcanic): from lava at surface. Intrusive: magma below surface.

Subject: Geology | Topic: Geology | Difficulty: easy

Q60.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C \cdot x$. Wait: $y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C \cdot x$. That's option A. Hmm, let me check: $y \, dx - x \, dy = 0 \Rightarrow y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow y = C \cdot x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q61.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = $1+4+9 = 14$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q62.

Answer: A

Solution:

$|\langle u, v \rangle| \leq \|u\| \cdot \|v\|$. Equality when u and v are linearly dependent.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q63.

Answer: C

Solution:

Rolle's: f continuous on $[a, b]$, differentiable on (a, b) , $f(a)=f(b)$. Then $f'(c)=0$ for some c in (a, b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q64.

Answer: B

Solution:

Extrusive (volcanic): from lava at surface. Intrusive: magma below surface.

Subject: Geology | Topic: Geology | Difficulty: moderate

Q65.

Answer: B

Solution:

Mohs: Talc(1)..Quartz(7)..Topaz(8)..Corundum(9)..Diamond(10). Diamond hardest.

Subject: Geology | Topic: Geology | Difficulty: difficult

PAPER BLUEPRINT

Subject Distribution:

Engineering Mathematics: 42 questions, 65 marks
Geology: 17 questions, 29 marks
Geophysics: 3 questions, 3 marks
General Aptitude: 1 questions, 1 marks
Remote Sensing & GIS: 1 questions, 1 marks
Petrology: 1 questions, 1 marks

Difficulty Distribution:

Easy: 26 | Moderate: 35 | Difficult: 4

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.